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Abhinav Joshi

PROFILE

Robotics engineer working on learning-based control, model-based optimal control, and full-stack autonomy across robot morphologies. On physical hardware, a full ROS 2 racing-autonomy stack on a 1:10 vehicle (EKF and particle-filter localization, SLAM, MPC on Jetson Orin Nano), monocular 6-DoF pose estimation replacing RGB-D on autonomous forklifts, real-time control-loop latency work on warehouse AMRs, and learned social navigation for a wheeled humanoid. In simulation, whole-body loco-manipulation (Go2 + SO-101, 18 DOF, single policy), humanoid whole-body RL and human-motion retargeting (Unitree G1, 29 DOF), floating-base optimal control in Crocoddyl and Pinocchio, and quadrotor RL and MPC. Also low-cost ultrasound- and IMU-driven assistive and rehabilitation control. Interested in unified autonomy across embodiments, world models for planning and control, and sim-to-real transfer.

EXPERIENCE

Robotics Intern — Mobile Robotics & Advanced Robotics Teams · Addverb Technologies — Jul 2026 – Present

- **Learned social navigation for Elexis-W**, India's first wheeled humanoid, a 133 kg differential-drive platform carrying two 7-DOF arms. Designed the reward formulation and curriculum, and trained the policy in Isaac Lab (detailed below). Sim-to-real to be tested on the physical humanoid.
- Next phase on the advanced robotics team: **dexterous tool manipulation**, building on the SimToolReal framework. Reliance-backed group working humanoid, quadruped and cobot systems.
- Developing **RL based perception blind whole body control** for **trakr quadruped** based on my Whole-Body Loco-Manipulation — Go2 + SO-101 (18 DOF, single RL policy) work.
- Solving **perception for autonomous mobile manipulation**, a **monocular 6-DoF/9-DoF pose-estimation pipeline for forklift pallet engagement**, replacing an RGB-D setup that fails on specular metal pallets and degrades under occlusion and changing illumination, while cutting sensor cost.
- Adapting **HCCEPose** (an open-source dense-correspondence method) to the pallet problem, with dense per-pixel 2D–3D correspondence, a Gray-coded coordinate representation, and a 4-way **PnP + RANSAC** hypothesis vote that resolves the 180° flip ambiguity and rejects glare-corrupted pixels by consensus. Built a **domain-randomized synthetic data pipeline** (BlenderProc → BOP) with deliberate degradation injection (specular lighting, occlusion, damage, loading), tuned for a low-cost onboard Intel compute target under a tight inference-latency budget.
- **Real-time control-loop latency on a tugger AMR.** The loop had to close at 50 Hz (20 ms) and was stuck near 200 ms. Software and driver were both clean; the composed communication path was the bottleneck. IPC over MODBUS/TCP → Ethernet switch → RTU-to-TCP gateway → serial RTU to the Italsea driver, two protocol conversions plus a low-baud serial leg with RTU's mandatory inter-frame idle gaps, stacking enough serialised delay to blow the budget. Bypassed the switch and the gateway, talking IPC to driver directly over serial at higher baud, collapsing the path back under 20 ms. Moving to **CAN bus** next for bounded latency as drives are added.
- **Low-level motor control** on an Italsea dual brushless permanent-magnet motor controller and a Moons' motor driver, over **MODBUS and CAN bus using the CiA 402** motion profile, the below-ROS layer that most policy work never touches.

Robotics Engineer & AI Researcher · RISE Lab, IIT Delhi — Nov 2025 – Present

- Developed and deployed a **full ROS 2 autonomy stack** for the F1Tenth / RoboRacer competition on a physical 1:10 vehicle (detailed below).
- Building a **low-cost ultrasound-driven rehabilitation and assistive system** for spinal-cord-injury patients. A-mode and B-mode sonomyography with IMU sensing decode muscle intent to drive a gamified rehab loop and an SO-101 assistive arm, keeping the patient in the loop.
- **Embedded and mechatronic design.** BNO085 and x-IMU3 sensing over ROS 2 and micro-ROS, digital twins in NVIDIA Isaac Sim with real-time sim-physical synchronisation, firmware on Teensy and ESP32.
- **SAM3D markerless motion capture** (RISE Lab × Sports Authority of India) for athlete biomechanics and injury prevention, and a demonstration-data source for imitation learning.
- Designed and delivered the **SO-101 robotics curriculum** (MuJoCo, LeRobot, ACT, Diffusion Policy) for the IIT Delhi Executive Programme; TA, Spring School in Sports Technology 2026.

Director — Manufacturing · EPI Engineering & Projects India Pvt. Ltd. — 2015 – Present

- Lead operations of an electro-mechanical manufacturing company supplying **Siemens, Airtel, and L&T**, from design through production; driving the firm's move toward applied and educational robotics.

SCADA Engineer · Areva T&D (now GE Grid Solutions) · + Asst. Manager, Union Bank of India (2013–15) — 2010 – 2015

- Built and maintained **real-time monitoring, fault-detection, and communication systems** for high-voltage grid substations under strict reliability requirements.

SELECTED PROJECTS

Learned Social Navigation — Elexis-W Wheeled Humanoid (India's first)

Addverb · 133 kg, differential-drive base + two 7-DOF arms, 26 links · Isaac Sim / Isaac Lab · PPO, 2048 envs · asymmetric actor-critic

Crossing a space occupied by moving people, framed as a timing problem rather than a geometry one, because **an obstacle's future position is not a function of its current position**. Two design consequences follow. A single-frame range scan is provably insufficient, since a person walking toward the robot and a wall produce identical observations, so the policy observes **relative position and relative velocity per agent**. And distance is the wrong cost variable, because a distance penalty is minimised by standing still, which is the frozen-robot failure that dominates the field; charging **time-to-collision** instead removes that attractor structurally rather than tuning it away. The safety envelope is a **0.79 m disc set by arm reach rather than by the wheelbase**, so the body plan defines the problem and the drive system does not. **Diagnostic work:** for 9M steps the policy reached the goal zero times, and the cause was found in the viewer rather than in the metrics. The bare traverse costs 9.5 s at 0.826 m/s against a 12 s episode, leaving 1.8 s of slack, less than any real dodge, so the policy was being penalised for evading correctly. An ablation confirmed the diagnosis: the same checkpoint reached the goal 45% of the time with the pedestrian removed and 4% with it active. Doubling the episode, with the discount factor raised in step to keep the terminal reward visible from step 0, took arrival from **0% to 89%**. Separately, removing reverse from the action space eliminated procrastination as a strategy, since freezing and reversing were the same tactic under discounting. **Stage 0 result: 89.4% arrival, 9.9% collision, 0.7% timeout, mean closest approach 2.46 s of time-to-collision against a 3 s horizon.** A useful negative result: slowing the pedestrian, intended to make evasion easier, made learning 6.5× worse, because corridor occupancy rather than closing speed is the binding constraint. Next: replacing privileged pedestrian state with onboard depth and RGB through teacher-student distillation, crowd density and reactivity curricula, and sim-to-real on the physical humanoid.

Autonomous Racing Stack — F1Tenth / RoboRacer

RoboRacer Competition · sole engineer · github.com/Abhinavjoshi7891/eth_roboracer

End-to-end **ROS 2 autonomy stack** on a physical 1:10 vehicle. **Model-based optimal control** in acados (both kinematic and single-track dynamic MPC with Pacejka tyres, real-time SQP) on a Jetson Orin Nano, a hybrid state estimator fusing IMU and VESC odometry with **particle-filter localization** and Cartographer SLAM in Frenet coordinates, a redesigned safety state machine for obstacle handling and overtaking, and full hardware bring-up (Hokuyo LiDAR, VESC FOC). Ported and re-architected the ForzaETH/TUM stack ROS 1 → ROS 2. Built a native **human-in-the-loop map and raceline editing stage inside the pipeline**, and replaced the fragile minimum-curvature velocity solve with an **O(N) two-pass friction-circle solver** that cannot ill-condition at small track scale. Tyre parameters fitted by nonlinear least squares from my own logged runs. Carried alone, including parts sourcing and customs into India.

Floating-Base Optimal Control — Crocoddyl / Pinocchio (Go2 + Talos)

FDDP · contact-mode switching · friction & wrench cones · warm-start · self-directed

Self-directed study of **whole-body optimal control on a floating base**, across a quadruped and a humanoid. Solved Go2 and Talos walking OCPs with **FDDP** (8.1× warm-start speedup on Talos), and built a **contact-mode switching experiment** comparing both robots. With horizon and costs fixed, changing only the support set moves optimal cost sharply, and Go2's single-leg support does not converge under a point-contact model where Talos's single-foot support does under a flat-sole wrench-cone model. The contact set, not leg count, is the discrete decision that determines achievable stability. Also ran Gepetto's CaT (constraints-as-terminations) on Solo12.

Aerial Optimal Control & RL — Quadrotor Policies + agilicious MPC

agilicious (UZH) · acados MPC · Aerial Gym · Isaac Gym / Isaac Lab · MuJoCo · PPO · RTX 4060 8 GB

Built and tested the **acados-based quadrotor MPC core** in agilicious, UZH's classical flight stack, and traced a solver-state corruption bug in its controller. Trained several **quadrotor RL policies** across three simulators, headline being **vision-based navigation through cluttered rooms from onboard depth** (VAE-compressed depth + GRU memory, async PPO) at 92% of the authors' RTX-4090 checkpoint on a laptop GPU. Wrote a **Crazyflie hover task from scratch** in MuJoCo. RL and control in simulation and on a research flight stack; preparing an autonomous drone-race hardware entry next.

Whole-Body Loco-Manipulation — Go2 + SO-101 (18 DOF, single RL policy)

Reproduction of Liu et al. (MLM, 2025) · Genesis + rsl_rl (PPO) · simulation, teleoperated live from a physical leader arm · RTX 4060 8 GB

Reproduced a 2025 whole-body loco-manipulation method on a merged Go2 + SO-101 platform (18 actuated DOF, one policy). Ported Isaac Gym → Genesis, built a procedural six-family 6-D trajectory library, and trained policies into a **live teleoperation system** driven by a physical SO-101 leader arm, gamepad, or keyboard: the robot walks, crouches, and stretches to track the operator's hand to **1–3 cm from history-only observation**. Extended the same platform to **bipedal locomotion** (rears onto hind legs, stands and walks at <1° mean pendulum angle) and to **agile quadrupedal skills** (running to 2.5 m/s, commanded jumping). A faithful reproduction, not an original method. **The quadruped work ran in simulation; the only physical hardware was the SO-101 leader arm.** Across eleven training runs every failure had the same shape, the policy correctly optimising the reward I wrote rather than the behaviour I meant, most sharply when a velocity reward computed in the body frame stopped being valid once the robot stood up. **Reward frames are part of morphology.**

Humanoid Whole-Body RL, MPC & Retargeting — Unitree G1 / Talos

Sim-to-Sim · PPO / FastSAC · Genesis + RSL-RL · Holosoma fork · MuJoCo Playground (JAX) · MJPC

Humanoid whole-body RL fork (**Holosoma**) optimised for consumer GPUs, extended with a **MediaPipe real-time retargeting pipeline** mapping live human motion to a 29-DOF G1. Trained a G1 joystick-walking policy (200M steps) in MuJoCo Playground, wrote a **sim2sim keyboard-teleop runner** loading the JAX/Orbax checkpoint live, and fixed two upstream bugs found during that work (a control-rate double-gating error, a brax null-config load crash). Built **MJPC** (google-deepmind/mujoco_mpc) from source and benchmarked live receding-horizon MPC on a humanoid stand/walk task, isolating a 4× realtime-factor gap to planner choice (Sampling vs. iLQG) rather than task difficulty, the same conclusion the Crocoddyl work above reaches from the opposite direction.

World Model for Model-Based Planning (DreamerV4-style)

JAX / Flax · space-time causal transformer · RSSM · SO-101 deployment in progress · 8 GB VRAM

Self-directed reproduction of a **DreamerV4-style latent world model** in JAX (space-time causal transformer, RSSM dynamics). **Reproduced and validated the VAE tokenizer on CoinRun**, compressing 64×64 frames into latent tokens and reconstructing them; the latent-imagination actor-critic loop is in progress. Now training a **small world model to deploy on a physical SO-101 arm**, to find out what breaks when a learned dynamics model meets real hardware. A reproduction in progress, not original methods or benchmarked results.

Dexterous Manipulation — LEAP Hand (MJX) & SimToolReal

MuJoCo Playground / MJX · Brax PPO · Isaac Sim 5.1 / Isaac Lab · training execution and codebase evaluation

Trained DeepMind's **LeapCubeReorient** task (16-DOF LEAP hand, freejoint cube, goal quaternion) to **205M timesteps** across 8192 parallel environments on an RTX 4060 laptop GPU, with two-stage checkpoint resumption, at 16–18k steps/s. Wrote an **interactive sim2sim runner** loading Brax PPO weights directly for live policy inspection in the MuJoCo viewer, the same tooling instinct that surfaced the Elexis-W episode-length bug. Separately configured and evaluated the open-source **SimToolReal** framework (Lum et al.) in Isaac Sim 5.1, running pre-trained LSTM-PPO / SAPG policies on a 29-DOF Kuka + SharPa hand across six tool tasks, in preparation for dexterous manipulation work at Addverb. Task, model, rewards and hyperparameters are the authors'; the training execution, tooling and evaluation are mine.

Open-Vocabulary 3D Scene Graphs for Language-Grounded Navigation

HOV-SG · SAM + OpenCLIP · RGB-D → 3D scene graph · in progress

Reproducing **HOV-SG** to turn RGB-D streams into a **language-queryable 3-D scene graph** (Building → Room → Object), so a robot can locate a target from a natural-language command. Paired with the world model above, the aim is a robot that plans over a structured 3-D scene from language. Reproduction and integration design in progress.

VLM-in-the-Loop Navigation (TortoiseBot)

ROS 2 · Vision-Language Navigation · Gemma 3 / Ollama

Language-instructed navigation on a differential-drive robot, sending multimodal payloads (camera, LiDAR, text) to a pretrained VLM for zero-shot control, with output clamping and safety-stop, plus a self-supervised dataset pipeline logging trajectories. An interactive demonstrator, not a deployed field system.

Quadruped Locomotion (RL) — Unitree Go2, Sim-to-Real

PPO · domain randomisation · Genesis + RSL-RL · consumer GPU

PPO locomotion for a Unitree Go2 with domain randomisation and sim-to-real transfer, including velocity- and height-command tracking and dynamic jumping behaviours, with a teleoperation runner and overlay tooling.

Occupancy-Grid Mapping & Map-Based Exploration

Teensy 4.1 · 3-wheel omni drive · ultrasonic sensing · built from scratch

Designed and built a **3-wheel holonomic robot** that maps an unknown space online into a 50×50 occupancy grid and **selects its own heading from that evolving map** via a direction-scoring routine over 24 sectors. Full holonomic kinematics, per-wheel PID, IMU heading correction.

Teleoperation Digital Twin — Franka Panda + Ability Hand

NVIDIA Isaac Sim · x-IMU3 arm tracking · 5DT glove / MediaPipe hands

Real-time pipeline mapping **human limb orientation and finger gestures** onto a Franka Panda + Ability Hand in Isaac Sim. Time-synchronised x-IMU3 sensors drive the arm, a 5DT glove or MediaPipe tracker drives the fingers, with a degraded-sensor fallback. URDF prep for Isaac (mimic joints → direct position drive).

Recursive Bayesian Localization — Clearpath Husky

LiDAR scan-to-map matching · GPS-free · unstructured environments · simulation

A recursive Bayesian estimator with LiDAR scan-to-map matching for GPS-free localization of a Clearpath Husky in unstructured environments, with explicit motion-model prediction and measurement-update steps.

TECHNICAL SKILLS

RL / LEARNING	Reinforcement Learning (PPO, SAC, FastSAC) Reward & Curriculum Design Asymmetric Actor-Critic Whole-Body Loco-Manipulation Teacher–Student / Privileged Learning Imitation Learning (ACT, Diffusion Policy) Domain Randomisation Sim-to-Real Transfer
OPTIMAL CONTROL	MPC (acados, CasADi) Kinematic & Single-Track Dynamic MPC Quadrotor MPC Sampling-Based MPC (MJPC) DDP / FDDP (Crocoddyl) Floating-Base Dynamics (Pinocchio) Contact / Friction & Wrench-Cone Constraints
ESTIMATION / PLANNING	EKF · Sensor Fusion Particle Filter / Recursive Bayesian SLAM (Cartographer, GMapping) 6-DoF Pose Estimation (PnP / RANSAC) Frenet-frame Planning Social & Dynamic-Obstacle Navigation
PERCEPTION / SIM	OpenCV · MediaPipe HCCEPose · BlenderProc / BOP SMPL-X / SAM3D Open-Vocab 3D (SAM + CLIP, HOV-SG) World Models (DreamerV4 / JAX) Isaac Sim / Isaac Lab MuJoCo / MJX / MJPC Genesis Aerial Gym
HRI / ASSISTIVE	Sonomyography (A/B-mode US) IMU / Biosignal Decoding Shared Autonomy Intent Estimation Teleoperation / Retargeting
SYSTEMS / HARDWARE	ROS 2 · ROS 1 · micro-ROS Jetson Orin Nano Teensy · STM32 · ESP32 LiDAR (Hokuyo) · VESC FOC Motor Drives (CiA 402) · MODBUS · CAN bus Python · C++ · MATLAB SolidWorks · Fusion 360 Linux · Docker · Git

EDUCATION

Executive Programme in Robotics · Top Performer 2025
IIT Delhi

M.Sc. Machine Learning & AI 2023
Liverpool John Moores University
Thesis: efficient sim-to-real drone navigation. Redesigned the student depth backbone (MobileNet-V3-Small, GhostNet) within a teacher–student framework

PG Diploma in ML & AI 2022
IIIT Bangalore

B.Tech Electrical & Electronics Engineering 2010
Bharati Vidyapeeth's College of Engineering

GRADUATE ADMITS (HELD)

M.Eng. Autonomy & Robotics — UIUC · **M.Eng. Robotics (Professional)** — University of Maryland

LANGUAGES & HONOURS

English — TOEFL 111/120, Duolingo 150/160 · **Advanced Drone Innovation**, IIT Hyderabad TiHAN 2024 · **RoboAI**, TOPAZ 2024 · **Silver Medalist**, National Boxing Championship

RESEARCH & PUBLICATIONS

EEG-Based Depression Detection: Hybrid CNN-LSTM with Spatial RGB Spectrogram Mapping
Pre-print, 2025 — Co-Author
InceptionV3 + LSTM with a novel spatial RGB spectrogram mapping for decoding structure from noisy neurophysiological signals. **90.0% LOSO accuracy (40 subjects); 93.8% MDD sensitivity.**

Ultrasound-Based Intent Decoding for Assistive & Rehabilitative Control
In preparation — target IEEE TBME / ICORR · RISE Lab, IIT Delhi
Sonomyography (A/B-mode ultrasound) for muscle-intent decoding driving a low-cost SO-101 arm for SCI rehabilitation and assistive control.

Teaching Drones to Navigate Through Urban Landscapes
M.Sc. Thesis, Liverpool John Moores University, 2023
Within a privileged-expert / student framework, redesigned the **student depth backbone** (MobileNet-V3-Small, GhostNet) to cut on-board compute while preserving **zero-shot obstacle avoidance**. Trained in simulation (Flightmare, Gazebo, Unity) with DAgger and domain randomization.